Energy revision sheet

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 State the main useful and wasted energy for an electric kettle.

- Useful: heat in the water. Wasted: heat to the surroundings and sound.

2 A motor uses 2000 J and gives 1500 J of kinetic energy. Calculate its efficiency.

- $\eta = 1500 \div 2000 = 0.75 = 75\%$

3 Give two ways of improving the efficiency of a machine with moving parts.

- Lubricate moving parts; use bearings; streamline to reduce air resistance.

4 State what an energy audit is used for.

- To list where energy is used and wasted in a system, so improvements can be made.